AL-NEELAIN UNIVERSITY

Faculty of Computer Science and Information Technology — Graduate Studies

Department of Computer Science | Artificial Intelligence Track

Course : COSC726 AGENTIC AI

المعايير المرجعية للرعاية السريرية لمرضى السكري — الجمعية الأمريكية للسكري (ADA 2024 / 2026)

American Diabetes Association (ADA) Standards of Care in Diabetes—2024 / 2026

[Comprehensive Clinical Reference & Master's Research Report (Diabetes Care)]

Researcher: Mohamed Abulgasim | Supervisor: Dr. Fakhreldeen Saeed | Date: 2026

Report on the American Diabetes Association (ADA)

1. Introduction

The American Diabetes Association (ADA) is one of the leading health and scientific organizations in the United States dedicated to combating diabetes. It works to prevent and treat diabetes through supporting scientific research, developing clinical guidelines, raising health awareness, and providing educational resources for patients, their families, and healthcare professionals.

Founded in 1940, the Association focuses on preventing diabetes, improving its treatment, enhancing the quality of life of people affected by the disease, and supporting scientific efforts aimed at developing more effective treatments, including the search for a cure.

The Association plays an important role in addressing the growing burden of diabetes and promoting evidence-based approaches to early detection, risk assessment, treatment, and long-term disease management.

2. Overview and Background of the Association

- **Official Name:** American Diabetes Association
- **Abbreviation:** ADA
- **Year of Establishment:** 1940
- **Headquarters:** Arlington, Virginia, United States
- **Official Website:** https://diabetes.org/

The Association operates through a network of volunteers, people living with diabetes, their families, healthcare providers, researchers, and employees. It also collaborates with scientific and healthcare organizations to promote diabetes prevention, improve diagnosis, and enhance treatment outcomes.

3. Vision and Mission of the Association

3.1 Vision

The Association's vision is to create a world where people can live free from the burdens of diabetes and its associated health, social, and economic consequences.

3.2 Mission

The ADA's mission focuses on three interconnected areas:

1. **Diabetes Prevention:** Supporting initiatives that reduce the risk of developing diabetes, particularly type 2 diabetes.
2. **Treatment and Scientific Research:** Funding and supporting research to improve diagnostic methods and treatments while pursuing a cure for diabetes.
3. **Improving Quality of Life:** Helping people with diabetes manage their condition and providing the information and resources they need to make informed health decisions.

The Association pursues its mission through scientific research, education and awareness, health advocacy, and efforts to improve access to appropriate healthcare services.

4. Main Objectives of the Association

The American Diabetes Association seeks to achieve several important health and scientific objectives, including:

1. **Supporting Scientific Research:** Funding studies investigating the causes of diabetes, its development, prevention, and treatment.
2. **Developing Clinical Guidelines:** Producing evidence-based recommendations to help healthcare professionals diagnose and manage diabetes.
3. **Promoting Health Awareness:** Providing information about nutrition, physical activity, and blood glucose monitoring.
4. **Improving Healthcare Quality:** Supporting access to appropriate medical services, medications, and healthcare technologies.

5. **Preventing Complications:** Helping reduce complications affecting the heart, kidneys, eyes, nerves, and feet.
6. **Advocating for Patients' Rights:** Supporting policies and initiatives that improve access to healthcare and reduce discrimination associated with diabetes.

5. Major Activities and Areas of Work

5.1 Scientific Research

The ADA supports research into the causes of diabetes, its underlying mechanisms, prevention strategies, and the development of medications, treatments, and technologies that improve patients' quality of life.

5.2 Education and Awareness

The Association provides educational resources for patients, families, and healthcare professionals. These resources cover diabetes management, blood glucose monitoring, nutrition, physical activity, and adherence to treatment plans.

5.3 Development of Clinical Guidelines

The ADA publishes the *Standards of Care in Diabetes* and updates them regularly to reflect new scientific evidence. These standards address diagnosis, treatment, glucose monitoring, and the prevention of diabetes-related complications.

5.4 Health Advocacy

The Association supports initiatives and policies intended to improve diabetes care, facilitate access to treatment and medications, and ensure that people affected by diabetes have access to reliable health information and resources.

These activities reflect the Association's central areas of work: research, advocacy, education, and the development of support networks.

6. Standards of Care in Diabetes

The *Standards of Care in Diabetes* represent one of the ADA's most significant contributions to diabetes management. They consist of evidence-based clinical recommendations designed to help healthcare professionals provide appropriate care for people with diabetes and those at risk of developing the disease.

The standards are updated annually and may also receive online updates during the year when new scientific evidence or clinical developments justify changes.

The main areas covered by the standards include:

Area	Main Topics
Diagnosis and Classification	Tests used to diagnose diabetes and identify its type
Prevention	Reducing the risk of developing type 2 diabetes
Glucose Monitoring	Measuring and tracking blood glucose levels
Pharmacological Treatment	Insulin and glucose-lowering medications
Nutrition and Physical Activity	Improving health behaviors and managing body weight
Cardiovascular and Kidney Health	Assessing risks and preventing complications
Eye, Nerve, and Foot Health	Early detection and management of complications
Special Populations	Children, older adults, and pregnant women
Hospital Care	Managing diabetes during hospitalization and medical procedures

Source: American Diabetes Association, *Standards of Care in Diabetes—2026*.

Key Updates in the 2026 Edition

The ADA released its 2026 Standards of Care in December 2025. Important updates include:

- Expanded recommendations concerning the use of Continuous Glucose Monitoring (CGM) devices.
- Updated guidance on glucose-lowering medications and obesity treatment for people with diabetes.
- Greater emphasis on the cardiovascular and kidney benefits of certain treatments.
- Updated nutritional recommendations addressing dietary patterns such as the Mediterranean diet and low-carbohydrate diets.
- Revised recommendations for diabetes management in children, older adults, and hospitalized patients.

The official standards are available at:

https://professional.diabetes.org/standards-of-care

7. The Association's Role in Diabetes Diagnosis

The ADA contributes to the development of clinical recommendations that help healthcare professionals identify diabetes through appropriate laboratory tests.

The main diagnostic tests include:

- **Hemoglobin A1c (HbA1c):** A blood test that reflects average blood glucose levels over the preceding months.
- **Fasting Plasma Glucose (FPG):** A blood test measuring plasma glucose after a period of fasting.
- **Oral Glucose Tolerance Test (OGTT):** A test used to evaluate the body's response to an oral glucose load.
- **Random Plasma Glucose:** A blood glucose measurement that can support diagnosis under specific clinical conditions.

The Association's standards define diagnostic thresholds for these tests. Diagnosis also depends on symptoms, clinical circumstances, and the need to confirm abnormal results when appropriate.

It is important to distinguish between **diagnostic criteria** and **treatment targets**, which are established after diagnosis and may vary according to the patient's circumstances.

8. The Association's Role in Preventing Diabetes Complications

Diabetes management involves more than controlling blood glucose levels. It also requires monitoring other risk factors that may affect a patient's overall health.

The ADA's clinical standards emphasize several important areas:

1. Monitoring blood pressure and blood lipid levels.
2. Assessing kidney function and detecting signs of kidney disease.
3. Conducting appropriate eye examinations to detect diabetic retinopathy.
4. Evaluating nerve function and foot health to reduce the risk of ulcers and injuries.
5. Providing nutritional education and encouraging appropriate physical activity.
6. Addressing mental health and the psychological challenges associated with living with diabetes.

Monitoring schedules and treatment targets vary according to diabetes type, age, overall health, and the medications being used.

9. The Importance of the ADA in Scientific Research and Artificial Intelligence Applications

The ADA's guidelines are particularly relevant to medical data analysis and machine learning projects because they provide a clinical reference for understanding diabetes-related variables and evaluating health-related outcomes.

Potential applications include:

- Developing machine learning models to predict the likelihood of type 2 diabetes.
- Analyzing factors associated with elevated blood glucose levels.
- Developing clinical decision-support systems using health data.
- Analyzing data collected by Continuous Glucose Monitoring devices.
- Evaluating predictive models using appropriate statistical and clinical performance metrics.

Practical Example: A Diabetes Prediction System

When developing a machine learning model using variables such as age, glucose level, Body Mass Index (BMI), and blood pressure, ADA guidelines can help researchers understand the clinical significance of these variables and interpret the model's results within an appropriate medical context.

However, several important considerations must be addressed:

- Training data alone cannot establish that a model is suitable for clinical use.
- The model should be evaluated using independent test data and appropriate metrics, such as sensitivity, specificity, and the Area Under the Receiver Operating Characteristic Curve (ROC-AUC).
- A model's prediction must not be treated as a definitive medical diagnosis.
- The model must be validated for the population in which it will be used, taking privacy, bias, and data quality into account.

Scientific Note: The ADA's clinical guidelines are a clinical reference, not a dataset for training an artificial intelligence model. Researchers should use suitable, reliable datasets and evaluate model performance against relevant clinical evidence and standards.

10. The Association's Resources and Scientific References

The following resources provide information about the ADA's activities and recommendations:

1. **Official ADA Website:** https://diabetes.org/
2. **Clinical Standards:** https://professional.diabetes.org/standards-of-care
3. **Diabetes Care Journal:** https://diabetesjournals.org/care
4. **Vision and Mission:** https://diabetes.org/about-us

These resources provide information on diabetes prevention, diagnosis, treatment, scientific research, and health education, as well as updated professional clinical guidelines.

11. Conclusion

The American Diabetes Association plays an important role in supporting scientific and medical efforts to reduce the burden of diabetes. It accomplishes this through research funding, health education, the development of evidence-based clinical standards, and efforts to improve access to appropriate healthcare services.

The Association is also an important scientific resource for researchers and developers working on diabetes prediction systems and patient data analysis. Nevertheless, applying its standards to technological solutions requires a clear understanding of the difference between statistical prediction and medical diagnosis, as well as rigorous evaluation of predictive models before they are used in clinical settings.

Therefore, the ADA is a valuable reference for academic studies and practical projects related to diabetes, particularly in the fields of digital health and artificial intelligence.

References

1. American Diabetes Association. (2026). *Standards of Care in Diabetes—2026*. Available at: https://professional.diabetes.org/standards-of-care
2. American Diabetes Association. *Our Vision and Mission*. Available at: https://diabetes.org/about-us
3. American Diabetes Association. (2025). *The American Diabetes Association Releases Standards of Care in Diabetes—2026*. Available at: https://diabetes.org/newsroom/press-releases/american-diabetes-association-releases-standards-care-diabetes-2026
4. American Diabetes Association Professional Practice Committee for Diabetes. (2026). *Diagnosis and Classification of Diabetes: Standards of Care in Diabetes—2026*. *Diabetes Care*. Available at: https://doi.org/10.2337/dc26-S002